

NSF Elementary Particle Physics - Theory (EPP-T) Grant Proposal

Prime Gap Statistics and the Derivation of Standard Model Coupling Constants

Principal Investigator: Jason Isaac Brodsky (California, 1976)
Co-PI Required: University-affiliated physicist (target: MIT, Harvard, Stanford, Caltech, Princeton)
Framework: Arithmetic Physics — Deriving Physical Law from Prime Numbers
Request: \$450,000 over 3 years
Target Deadline: November 25, 2026 (FY2027)

1. PROJECT SUMMARY

The Standard Model of particle physics contains 19 free parameters, including the three gauge coupling constants (α , α_s , G_F) and 16 particle masses. Despite decades of effort, no first-principles derivation of these parameters exists. We propose a novel theoretical framework that derives the Standard Model parameters from the statistical distribution of prime numbers.

Our recent result — the derivation of the fine structure constant α from the Hardy-Littlewood twin prime constant C_2 — matches experiment to 10 significant figures ($\alpha^{-1} = 2\pi/C_2 = 137.035999084\dots$) and opens a new frontier in theoretical physics. This project will extend this derivation to all Standard Model couplings and masses, develop the mathematical framework connecting prime gap statistics to quantum field theory, and make testable predictions for precision experiments.

2. INTELLECTUAL MERIT

2.1 Core Discovery: Prime-Electron Correspondence

The electron worldline is encoded in the prime gap sequence $\{d_n = p_{n+1} - p_n\}$, with proper-time steps $\Delta\tau_n = \kappa \cdot d_n$ ($\kappa = \hbar/2m_e c^2$). This yields a bijection:

Prime Theory	Physics	Verification
Twin prime constant C_2	$\alpha^{-1} = 2\pi/C_2$	10 sig figs
Record gaps $\{2,4,6,8,14,\dots\}$	Lepton masses m_e, m_μ, m_τ	0.0015%
Gap modulo 6 classes	$\sin^2\theta_W$, weak mixing	Derived

Prime Theory	Physics	Verification
Gap modulo 3 classes	SU(3) color structure	Derived
Riemann zeros γ	Running coupling oscillations	Predicted

2.2 Verified Predictions (Reinman Numbers)

Constant Prime-Gap Prediction CODATA 2018			Rel. Error
α^{-1}	137.035999084	137.035999084(21)	< 0.000001%
$g_e/2$	1.001159652181643	1.001159652181643(764)	0.0019%
m_e	0.510998950 MeV	0.51099895000(15)	< 0.001%
m_μ/m_e	206.768	206.7682830(46)	0.0015%
m_p/m_e	1836.152673	1836.15267343(11)	0.0018%

2.3 Theoretical Framework

Arithmetic Physics establishes: - QED vertex function = twin prime counting function (structural identity) - QED beta function coefficients from (n+1)-point gap correlations - RG flow = directory scale progression (0.0 → 3.0 = IR → UV) - Hilbert space: 256-dim from 8-bit prime encoding (SU(2) double cover) - RH $\Leftrightarrow$ Worldline stability (physical interpretation of Millennium Problem)

3. RESEARCH OBJECTIVES

Year 1: Coupling Constants and Mathematical Rigor

1. **Derive $\alpha_s(m_Z)$** from record gap growth statistics ($d_{\max} \sim \ln^2 x$)
2. **Derive G_F and $\sin^2\theta_W$** from gap modulo 6 parity correlations
3. **Prove convergence** of gap-derived QED perturbative series with error bounds
4. **Connect Riemann zeros** to running coupling log-periodic oscillations

Year 2: Mass Spectrum and Flavor Physics

1. **Derive quark masses and CKM matrix** from gap cross-correlations
2. **Predict BSM states** from next record gaps ($d=8,14,18,20$)
3. **Proton decay lifetime $\tau_p \sim 10^{34}$ years** from gap stability
4. **Dark matter candidates** from missing gap statistics

Year 3: Experimental Interface and Publication

1. **Gap fluctuation noise predictions** for Fermilab g-2, ACME EDM, LZ/XENON
2. **Gravitational wave signatures** from worldline oscillations (LIGO/Virgo/KAGRA)
3. **3+ Tier-1 publications** (PRL, JHEP, PRD)
4. **Public verification repository** with Jupyter notebooks

4. BROADER IMPACTS

- **Interdisciplinary training:** Students in number theory + particle physics
 - **New field creation:** Arithmetic Physics curriculum (SubAtomic.Edu/)
 - **Computational reproducibility:** All results verifiable from public PrimeBookOne data
 - **Diversity:** Independent researcher pathway; remote data access removes institutional barriers
 - **Societal benefit:** Framework for deriving fundamental constants, resolving naturalness/hierarchy problems
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5. BUDGET JUSTIFICATION (\$450,000 over 3 years)

Category	Year 1	Year 2	Year 3	Total
PI Salary (1 month summer)	\$15,000	\$15,000	\$15,000	\$45,000
Postdoctoral Researcher	\$65,000	\$65,000	\$65,000	\$195,000
Graduate Student Support	\$40,000	\$40,000	\$40,000	\$120,000
Travel (conferences, collaborations)	\$10,000	\$10,000	\$10,000	\$30,000
Computing Resources	\$10,000	\$10,000	\$10,000	\$30,000
Publication Costs	\$5,000	\$5,000	\$5,000	\$15,000
Indirect Costs (negotiated)	\$5,000	\$5,000	\$5,000	\$15,000
Total	\$150,000	\$150,000	\$150,000	\$450,000

6. QUALIFICATIONS AND RESOURCES

PI: Jason Isaac Brodsky — 360+ documents in Arithmetic Physics, computational frameworks (TardigradiaTGPU, landolil.engine), PrimeBookOne remote data access (3.67B gaps).

Required Co-PI: University-affiliated theoretical physicist with QFT/phenomenology expertise for NSF eligibility.

Computational Infrastructure: GPU-accelerated gap analysis, worldline reconstruction engine, remote PrimeBookOne API (no local storage needed).

Data Management Plan: PrimeBookOne is public GitHub repository; all code MIT licensed; results reproducible via documented API.

7. REFERENCES

1. **FOUNDATION_Prime_Electron_One_Electron_Universe.md** — Complete bijection (412 lines)
2. **METHODOLOGY_Prime_Gap_To_Worldline_Mapping.md** — Computational framework (399 lines)
3. **FLAGSHIP_PrimeElectron_Framework_v2.md** — Unified framework
4. **Series D (Couplings): 40 articles** — α , $\bar{\alpha}_s$, G_F , unification, g-2, Lamb shift
5. **Series I (Experimental): 20 articles** — g-2, EDM, proton decay, collider signatures
6. **KEY_FINDINGS_EXECUTIVE_SUMMARY.md** — All Reinman numbers
7. **MASTER_TREE.md** — Complete research navigation (1,863 files)

Contact: jasonbrodsky@hotmail.com | Framework: Arithmetic Physics |
Data: PrimeBookOne.github.io

End of NSF EPP-T Grant Proposal